SMART INDIA HACKATHON 2026 • TECHNICAL ARCHITECTURE

Enterprise Student Experience Blueprint

Student Experience: Automated Resume Onboarding, AI Certificate Verification & Core Features

Complete Non-Breaking Engineering Specification: Seamless First-Login Journey • Zero Student API Key Dependency • Multi-Stage Certificate Credibility Engine

Platform:	Privacy Standard:	Verification Tech:	Architecture:
SkillSetu Core Enterprise	100% Server-Side AI (No Student Keys)	Multimodal Vision AI + QR Resolver	4-Phase Safe Non-Breaking Pipeline

🔒 **Architecture Decision: Zero-Trust Client Model (No Student API Keys Ever)**

Students are

never asked for personal API keys

. Requiring students to supply private keys damages trust and risks accidental leakage. All AI telemetry (Resume Ingestion, Skill Mapping, Course Scouting, and Certificate Verification) runs

100% on the platform's secure server infrastructure

using managed Gemini 2.5 Flash and local Python processors. The student experiences instant, frictionless automation at zero cost.

1. The Seamless 5-Step First-Login Onboarding Lifecycle

User Journey

How a new student transitions from initial login to verified dashboard readiness:

1. **Initial Login** — Student logs in via college credentials or OAuth. System detects first-time user flag (`is_onboarded: false`).
2. **Immediate Resume Ingestion** — System prompts drag-and-drop resume upload (PDF). Local server parses text in <100ms with DPDP PII masking.
3. **AI Profile Synthesis** — Server AI synthesizes preliminary profile: extracts skills, education, GitHub repos, and initial proficiency levels.
4. **Interactive Review & Verification** — Student reviews profile: adds missing achievements, projects, and uploads certificates for AI credibility audit.
5. **Landing Page Launch** — Student is redirected to Dashboard with populated match rings, verified badges, and personalized opportunities!

2. Dedicated AI Certificate Credibility & Fraud Verification Engine

Anti-Fraud Technology

Specialized AI Tech Stack for Authentic Certificate Verification

Tamper-Detection Engine

- **The Three-Layer Credibility Pipeline**

When a student uploads a course or achievement certificate (PDF or Image), the system passes it through a 3-layer verification pipeline before adding it to their verified profile:

1. **Layer 1: Multimodal Vision AI (Gemini 2.5 Flash Vision / LayoutLMv3):** Analyzes the document image for official issuing authority typography, recognized logo vectors, standardized course codes, and authorized digital signatures (e.g., NPTEL, Coursera, AWS, edX, IITs).
2. **Layer 2: PDF XMP Metadata & Splicing Anomaly Detection (Python `pikepdf` & OpenCV):** Inspects the binary metadata of the PDF. Flags documents modified by image manipulation software (e.g., Adobe Photoshop, Canva, Illustrator), detects mismatched font layers (where a student edits the name over someone else's certificate), and catches altered date strings.
3. **Layer 3: Automated QR Code & Registry Resolver (Python `pyzbar` + Registry Webhooks):** Automatically decodes embedded QR codes or verification URLs (e.g., `nptel.ac.in/noc/Ecertificate` or `coursera.org/verify/XXXX`). Queries the issuing academy's public verification endpoint to confirm matching student name, course title, and completion grade.

- **Verification Outcome & TrustLedger Minting**

Confidence Score	Status Assigned	Platform Action
≥ 90% Match	VERIFIED	Displays verified green check badge on student's public portfolio; mints SHA-256 certificate block to TrustLedger.
70% – 89% Match	PENDING_AUDIT	Added to profile as unverified self-claim until secondary manual or automated registry webhook confirmation.
< 70% / Tampered	REJECTED / FORGED	Certificate rejected with clear explanation: "*Document failed metadata integrity check (Font layer splice detected).*"

3. Step-by-Step Engineering Blueprints for Core Student Features

Developer Guide

Feature 1: First-Login Resume Ingestion & Profile Synthesis Wizard

Onboarding Wizard • Route: /student/onboarding

▪ **How to Build It Without Disrupting the Existing App**

- **Trigger:** Inside `AuthContext.jsx`, after successful login, inspect `user.onboarded`. If false, route the user to `/student/onboarding` instead of the main dashboard.
- **Upload Interface:** A clean 3-step modal wizard: *Step 1: Upload Resume • Step 2: AI Extracts Profile • Step 3: Review & Add Certificates/Projects • Finish → Redirect to Dashboard.*
- **Backend Controller:** `POST /api/student/onboard/parse-resume` extracts raw text via `pypdf`, sends sanitized prompt to Gemini, and returns structured JSON (skills array, degree, CGPA, graduation year, GitHub URL).
- **Fallback:** If the student skips resume upload, allow them to enter basic details manually without blocking access.

Feature 2: Autonomous Server-Side AI Course Scout & Budget Tiers

Continuous Upskilling • Route: /student/learning

▪ **How to Build It (100% Server-Side • No Student Keys)**

- **Server Background Worker:** A scheduled asynchronous task (via Python `APScheduler` or FastAPI background tasks) runs periodically to check accredited sources (NPTEL, SWAYAM, MIT OCW, Coursera, AWS Skill Builder) for new courses and updated syllabus versions.
- **Smart Course Matching:** Evaluates student skill gaps against newly scouted courses on the server.
- **Budget Filter UI:** In `Learning.jsx`, display three simple tabs: **100% Free / Govt-Funded (₹0) • Micro-Boosters (<₹1,000) • Sponsored Grants**.
- **Zero Clutter:** Shows cards in clean 3-column responsive grid with official academy logos and direct enrollment links.

Feature 3: AI Voice & Audio/Video Mock Technical Interview Simulator

Skill Preparedness • Route: /student/mock-interview

▪ **How to Build It Using Standard Web APIs**

- **Client-Side Speech Processing:** Uses standard browser `window.SpeechRecognition` for real-time speech-to-text and `window.speechSynthesis` for spoken AI questions. Zero external audio latency.
- **Evaluation Engine:** `POST /api/interview/evaluate` sends transcribed student answers to server Gemini AI, evaluating Technical Accuracy (50%), Keyword Coverage (25%), and Communication Clarity (25%).
- **UI Layout:** Standalone distraction-free route with dark studio styling, audio wave animation, and camera HUD widget for posture and eye-contact feedback.

Feature 4: Dynamic ATS Resume Scorer & 1-Click Keyword Tailor

Job Fit Optimizer • Inside Opportunities Modal

▪ **How to Build It Without Crowding Opportunities Page**

- **Slide-Over Drawer:** An *"Audit Resume for this Role"* button inside the opportunity detail modal opens an elegant slide-over drawer without navigating away from the job.
- **Instant Feedback:** Displays an animated ATS score ring (e.g. 78%) and two clear columns: "*Matching Keywords in Resume*" vs. "*Missing Keywords to Include*".
- **Privacy:** Scans PDF in memory, runs PII redaction, and deletes temporary files immediately after evaluation.

Feature 5: Hackathon Teammate Matcher & Micro-Internship Bounties

Community & Hands-On Experience

▪ **How to Build It as Lightweight Modular Views**

- **Teammate Matcher (`/student/teams`):** Compact card layout where students post project ideas with required complementary skills (e.g. Frontend React + Backend Python + Cloud DevOps).
- **Micro-Bounties Toggle:** Embedded directly inside `Opportunities.jsx` as a simple toggle: **[Full Internships | 2-Week Micro-Bounties]**. Bounties link to GitHub pull requests and award verified badges on TrustLedger.

4. The 4-Phase Safe Implementation Timeline & Testing Checklist

Engineering Delivery

Phase 1: Schemas, OCR & Router Stubs

Week 1

Build independent database tables (`certificates`, `dynamic_courses`, `bounties`) and FastAPI router stubs. Implement local PDF text parser and QR code resolver. Test endpoints in Swagger UI (`/api/docs`).

Phase 2: Onboarding Wizard & UI Shells

Week 2

Create `OnboardingWizard.jsx` and register routes in `App.jsx`. Build the Certificate Upload modal with drag-and-drop support and the ATS Resume Audit drawer. Ensure zero layout regressions on the existing dashboard.

Phase 3: AI Intelligence & Vision Pipeline

Week 3

Connect server-side Gemini 2.5 Flash for resume profile synthesis, certificate vision analysis, and mock interview scoring. Implement PDF metadata inspection to flag manipulated/forged documents.

Phase 4: TrustLedger Minting & Hardening

Week 4

Anchor verified certificate hashes and proctored interview scores to the sovereign TrustLedger. Conduct end-to-end responsive audits across mobile and desktop. Verify zero console warnings.

Developer Testing & Production Safety Verification

Sanity Checklist

Test Scenario	Verification Action	Expected Safe Behavior
Onboarding Skip / Failure	Upload an unreadable or corrupt resume PDF file during initial onboarding wizard.	UI informs user gently: "*Couldn't parse PDF, please fill basic info*"; does not throw React error.
Forged Certificate Test	Upload an image with spliced/edited text layer or Photoshop metadata.	Metadata inspector detects editing tool; status marked as `PENDING_AUDIT` or `REJECTED`.
Server AI Timeout	Simulate AI API rate-limit or temporary network drop during course scouting.	UI gracefully displays cached accredited course catalog with zero screen flickering.
Mobile Viewport Audit	Test onboarding wizard, resume upload, and interview simulator on 375px mobile screen.	Modals resize smoothly, buttons remain fully accessible, and camera HUD scales cleanly.

💡 **Presentation Defense for SIH Judges:**

"Judges, our student onboarding is effortless and cryptographically secure. The moment a student logs in, our server-side AI analyzes their resume, populates their profile, and verifies their certificates using a

Multimodal Vision & Metadata Anti-Forgery Engine

. Students never need to expose personal API keys, and recruiters are guaranteed that every listed certificate on the platform is genuine and verified on our

Sovereign SHA-256 TrustLedger

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